

EEET202

HINES & TRANSFORMERS

MODULE: 4

– All Day Efficiency of Transformer

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Syllabus

ers –constructional details, principle of operation, EMF equation, convention, magnetising current, transformation ratio, phasor no load and on load, equivalent circuit, percentage and per unit regulation. Transformer losses and efficiency, condition for 7/A rating. Testing of transformers– polarity test, open circuit test, Sumpner's test – separation of losses, all day efficiency.Parallel transformers– numerical problems

All-day Efficiency

s used in distribution systems, have their primary
sed all the 24 hours, although their secondary
little or no-load much of the time during the day.
core losses occur throughout the day whereas
cur only when transformer is loaded.

e of such a transformer is judged by all-day
n the basis of energy consumed during a certain
ally a day of 24 hours.

$$\text{All-day Efficiency} = \frac{\text{Output in KWh}}{\text{Input in KWh}}$$

Problem

efficiency of 500 KVA distribution transformer and iron loss at full load are 4.5 KW and 3.5 KW a day of 24 hours, it is loaded as under:

Loading in KW Power factor

0.8

0.75

0.8

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$$\frac{\text{Input in KWhr}}{\text{Output in KWhr}} = \frac{\text{Output in KWhr}}{\text{Output in KWhr} + \text{Losses in KWhr}}$$

$$\text{hr} = 6 \times 400 + 10 \times 300 + 4 \times 100$$

$$= 2400 + 3000 + 400$$

$$= \underline{\underline{5800 \text{ KWhr}}}$$

ly with load condition.

$$\text{cu loss} = 4.5 \text{ kW.}$$

$$\text{at } 0.8 \text{ pf} = \frac{400}{0.8} \text{ kVA}$$

$$= 500 \text{ kVA} \Rightarrow \text{full load.}$$

$$\text{at } 0.75 \text{ pf} = 400 \text{ kVA} \Rightarrow 80\% \text{ full load}$$

$$\text{at } 0.8 \text{ pf} = 125 \text{ kVA} \Rightarrow 25\% \text{ full load.}$$

$$I^2 r_{01}$$

$$\begin{aligned} x \text{ times full load} &= (x I)^2 r_{01} \\ &= x^2 I^2 r_{01} \end{aligned}$$

$$= x^2 (\text{full-load cu. loss}).$$

$$0 \text{ kW}, 0.8 \text{ pf} = 40.5 \text{ kW}$$

$$\begin{aligned} &= 40.5 \times 6 \text{ kWh} \\ &= 24 \text{ kWh} \end{aligned}$$

$$\begin{aligned} 4, 0.75 \text{ pf} &= 0.8^2 \times 40.5 = 2.88 \text{ kW} \\ &= 28.8 \text{ kWh} \end{aligned}$$

$$\begin{aligned} 1, 0.8 \text{ pf} &= 0.85^2 \times 40.5 = 0.2812 \text{ kW} \\ &= 1.25 \text{ kWh} \end{aligned}$$

$$s \text{ at km/hr} = 56.925 \text{ km/hr}$$

$$= \frac{5800}{5800 + 84 + 56.925}$$

$$= 0.9763$$

$$= 97.63\%$$

$$\underline{\underline{= 97.63\%}}$$